Distributed Systems

Theory and Applications

Ratan K. Ghosh • Hiranmay Ghosh



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Distributed Systems

Theory and Applications

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Contents

About the Authors *xv*
Preface *xvii*
Acknowledgments *xxi*
Acronyms *xxiii*

1 Introduction *1*
1.1 Advantages of Distributed Systems *1*
1.2 Defining Distributed Systems *3*
1.3 Challenges of a Distributed System *5*
1.4 Goals of Distributed System *6*
1.4.1 Single System View *7*
1.4.2 Hiding Distributions *7*
1.4.3 Degrees and Distribution of Hiding *9*
1.4.4 Interoperability *10*
1.4.5 Dynamic Reconfiguration *10*
1.5 Architectural Organization *11*
1.6 Organization of the Book *12*
Bibliography *13*

2 The Internet *15*
2.1 Origin and Organization *15*
2.1.1 ISPs and the Topology of the Internet *17*
2.2 Addressing the Nodes *17*
2.3 Network Connection Protocol *20*
2.3.1 IP Protocol *22*
2.3.2 Transmission Control Protocol *22*
2.3.3 User Datagram Protocol *22*
2.4 Dynamic Host Control Protocol *23*
2.5 Domain Name Service *24*

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2.5.1 Reverse DNS Lookup *27*
2.5.2 Client Server Architecture *30*
2.6 Content Distribution Network *32*
2.7 Conclusion *34*
Exercises *34*
Bibliography *35*

3 Process to Process Communication *37*
3.1 Communication Types and Interfaces *38*
3.1.1 Sequential Type *38*
3.1.2 Declarative Type *39*
3.1.3 Shared States *40*
3.1.4 Message Passing *41*
3.1.5 Communication Interfaces *41*
3.2 Socket Programming *42*
3.2.1 Socket Data Structures *43*
3.2.2 Socket Calls *44*
3.3 Remote Procedure Call *48*
3.3.1 XML RPC *52*
3.4 Remote Method Invocation *55*
3.5 Conclusion *59*
Exercises *59*
Additional Web Resources *61*
Bibliography *61*

4 Microservices, Containerization, and MPI *63*
4.1 Microservice Architecture *64*
4.2 REST Requests and APIs *66*
4.2.1 Weather Data Using REST API *67*
4.3 Cross Platform Applications *68*
4.4 Message Passing Interface *78*
4.4.1 Process Communication Models *78*
4.4.2 Programming with MPI *81*
4.5 Conclusion *87*
Exercises *88*
Additional Internet Resources *89*
Bibliography *89*

5 Clock Synchronization and Event Ordering *91*
5.1 The Notion of Clock Time *92*
5.2 External Clock Based Mechanisms *93*

10.1002/9781119823963.fmatter, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9781119823963.fmatter by Readcube (Labtiva Inc.), Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

5.2.1 Cristian's Algorithm *93*
5.2.2 Berkeley Clock Protocol *94*
5.2.3 Network Time Protocol *95*
5.2.3.1 Symmetric Mode of Operation *96*
5.3 Events and Temporal Ordering *97*
5.3.1 Causal Dependency *99*
5.4 Logical Clock *99*
5.5 Causal Ordering of Messages *106*
5.6 Multicast Message Ordering *107*
5.6.1 Implementing FIFO Multicast *110*
5.6.2 Implementing Causal Ordering *112*
5.6.3 Implementing Total Ordering *113*
5.6.4 Reliable Multicast *114*
5.7 Interval Events *115*
5.7.1 Conceptual Neighborhood *116*
5.7.2 Spatial Events *118*
5.8 Conclusion *120*
Exercises *121*
Bibliography *123*

6 Global States and Termination Detection *127*
6.1 Cuts and Global States *127*
6.1.1 Global States *132*
6.1.2 Recording of Global States *134*
6.1.3 Problem in Recording Global State *138*
6.2 Liveness and Safety *140*
6.3 Termination Detection *143*
6.3.1 Snapshot Based Termination Detection *144*
6.3.2 Ring Method *145*
6.3.3 Tree Method *148*
6.3.4 Weight Throwing Method *151*
6.4 Conclusion *153*
Exercises *154*
Bibliography *156*

7 Leader Election *157*
7.1 Impossibility Result *158*
7.2 Bully Algorithm *159*
7.3 Ring-Based Algorithms *160*
7.3.1 Circulate IDs All the Way *161*
7.3.2 As Far as an ID Can Go *162*

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7.4 Hirschberg and Sinclair Algorithm *163*
7.5 Distributed Spanning Tree Algorithm *167*
7.5.1 Single Initiator Spanning Tree *167*
7.5.2 Multiple Initiators Spanning Tree *170*
7.5.3 Minimum Spanning Tree *176*
7.6 Leader Election in Trees *176*
7.6.1 Overview of the Algorithm *176*
7.6.2 Activation Stage *177*
7.6.3 Saturation Stage *178*
7.6.4 Resolution Stage *179*
7.6.5 Two Nodes Enter SATURATED State *180*
7.7 Leased Leader Election *182*
7.8 Conclusion *184*
Exercises *185*
Bibliography *187*

8 Mutual Exclusion *189*
8.1 System Model *190*
8.2 Coordinator-Based Solution *192*
8.3 Assertion-Based Solutions *192*
8.3.1 Lamport's Algorithm *192*
8.3.2 Improvement to Lamport's Algorithm *195*
8.3.3 Quorum-Based Algorithms *196*
8.4 Token-Based Solutions *203*
8.4.1 Suzuki and Kasami's Algorithm *203*
8.4.2 Singhal's Heuristically Aided Algorithm *206*
8.4.3 Raymond's Tree-Based Algorithm *212*
8.5 Conclusion *214*
Exercises *215*
Bibliography *216*

9 Agreements and Consensus *219*
9.1 System Model *220*
9.1.1 Failures in Distributed System *221*
9.1.2 Problem Definition *222*
9.1.3 Agreement Problem and Its Equivalence *223*
9.2 Byzantine General Problem (BGP) *225*
9.2.1 BGP Solution Using Oral Messages *228*
9.2.2 Phase King Algorithm *232*
9.3 Commit Protocols *233*
9.3.1 Two-Phase Commit Protocol *234*

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9.3.2 Three-Phase Commit *238*
9.4 Consensus *239*
9.4.1 Consensus in Synchronous Systems *239*
9.4.2 Consensus in Asynchronous Systems *241*
9.4.3 Paxos Algorithm *242*
9.4.4 Raft Algorithm *244*
9.4.5 Leader Election *246*
9.5 Conclusion *248*
Exercises *249*
Bibliography *250*

10 Gossip Protocols *253*
10.1 Direct Mail *254*
10.2 Generic Gossip Protocol *255*
10.3 Anti-entropy *256*
10.3.1 Push-Based Anti-Entropy *257*
10.3.2 Pull-Based Anti-Entropy *258*
10.3.3 Hybrid Anti-Entropy *260*
10.3.4 Control and Propagation in Anti-Entropy *260*
10.4 Rumor-mongering Gossip *261*
10.4.1 Analysis of Rumor Mongering *262*
10.4.2 Fault-Tolerance *265*
10.5 Implementation Issues *265*
10.5.1 Network-Related Issues *266*
10.6 Applications of Gossip *267*
10.6.1 Peer Sampling *267*
10.6.2 Failure Detectors *270*
10.6.3 Distributed Social Networking *271*
10.7 Gossip in IoT Communication *273*
10.7.1 Context-Aware Gossip *273*
10.7.2 Flow-Aware Gossip *274*
10.7.2.1 Fire Fly Gossip *274*
10.7.2.2 Trickle *275*
10.8 Conclusion *278*
Exercises *279*
Bibliography *280*

11 Message Diffusion Using Publish and Subscribe *283*
11.1 Publish and Subscribe Paradigm *284*
11.1.1 Broker Network *285*
11.2 Filters and Notifications *287*

10.1002/9781119823963.fmatter, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9781119823963.fmatter by Readcube (Labtiva Inc.), Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

11.2.1 Subscription and Advertisement *288*
11.2.2 Covering Relation *288*
11.2.3 Merging Filters *290*
11.2.4 Algorithms *291*
11.3 Notification Service *294*
11.3.1 Siena *294*
11.3.2 Rebeca *295*
11.3.3 Routing of Notification *296*
11.4 MQTT *297*
11.5 Advanced Message Queuing Protocol *299*
11.6 Effects of Technology on Performance *301*
11.7 Conclusions *303*
Exercises *304*
Bibliography *305*

12 Peer-to-Peer Systems *309*
12.1 The Origin and the Definition of P2P *310*
12.2 P2P Models *311*
12.2.1 Routing in P2P Network *312*
12.3 Chord Overlay *313*
12.4 Pastry *321*
12.5 CAN *325*
12.6 Kademlia *327*
12.7 Conclusion *331*
Exercises *332*
Bibliography *333*

13 Distributed Shared Memory *337*
13.1 Multicore and S-DSM *338*
13.1.1 Coherency by Delegation to a Central Server *339*
13.2 Manycore Systems and S-DSM *340*
13.3 Programming Abstractions *341*
13.3.1 MapReduce *341*
13.3.2 OpenMP *343*
13.3.3 Merging Publish and Subscribe with DSM *345*
13.4 Memory Consistency Models *347*
13.4.1 Sequential Consistency *349*
13.4.2 Linearizability or Atomic Consistency *351*
13.4.3 Relaxed Consistency Models *352*
13.4.3.1 Release Consistency *356*
13.4.4 Comparison of Memory Models *357*

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13.5 DSM Access Algorithms *358*
13.5.1 Central Sever Algorithm *359*
13.5.2 Migration Algorithm *360*
13.5.3 Read Replication Algorithm *361*
13.5.4 Full Replication Algorithm *362*
13.6 Conclusion *364*
Exercises *364*
Bibliography *367*

14 Distributed Data Management *371*
14.1 Distributed Storage Systems *372*
14.1.1 RAID *372*
14.1.2 Storage Area Networks *372*
14.1.3 Cloud Storage *373*
14.2 Distributed File Systems *375*
14.3 Distributed Index *376*
14.4 NoSQL Databases *377*
14.4.1 Key-Value and Document Databases *378*
14.4.1.1 MapReduce Algorithm *380*
14.4.2 Wide Column Databases *381*
14.4.3 Graph Databases *382*
14.4.3.1 Pregel Algorithm *384*
14.5 Distributed Data Analytics *386*
14.5.1 Distributed Clustering Algorithms *388*
14.5.1.1 Distributed K-Means Clustering Algorithm *388*
14.5.2 Stream Clustering *391*
14.5.2.1 BIRCH Algorithm *392*
14.6 Conclusion *393*
Exercises *394*
Bibliography *395*

15 Distributed Knowledge Management *399*
15.1 Distributed Knowledge *400*
15.2 Distributed Knowledge Representation *401*
15.2.1 Resource Description Framework (RDF) *401*
15.2.2 Web Ontology Language (OWL) *406*
15.3 Linked Data *407*
15.3.1 Friend of a Friend *407*
15.3.2 DBpedia *408*
15.4 Querying Distributed Knowledge *409*
15.4.1 SPARQL Query Language *410*

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15.4.2 SPARQL Query Semantics *411*
15.4.3 SPARQL Query Processing *413*
15.4.4 Distributed SPARQL Query Processing *414*
15.4.5 Federated and Peer-to-Peer SPARQL Query Processing *416*
15.5 Data Integration in Distributed Sensor Networks *421*
15.5.1 Semantic Data Integration *422*
15.5.2 Data Integration in Constrained Systems *424*
15.6 Conclusion *427*
Exercises *428*
Bibliography *429*

16 Distributed Intelligence *433*
16.1 Agents and Multi-Agent Systems *434*
16.1.1 Agent Embodiment *436*
16.1.2 Mobile Agents *436*
16.1.3 Multi-Agent Systems *437*
16.2 Communication in Agent-Based Systems *438*
16.2.1 Agent Communication Protocols *439*
16.2.2 Interaction Protocols *440*
16.2.2.1 Request Interaction Protocol *441*
16.3 Agent Middleware *441*
16.3.1 FIPA Reference Model *442*
16.3.2 FIPA Compliant Middleware *443*
16.3.2.1 JADE: Java Agent Development Environment *443*
16.3.2.2 MobileC *443*
16.3.3 Agent Migration *444*
16.4 Agent Coordination *445*
16.4.1 Planning *447*
16.4.1.1 Distributed Planning Paradigms *447*
16.4.1.2 Distributed Plan Representation and Execution *448*
16.4.2 Task Allocation *450*
16.4.2.1 Contract-Net Protocol *450*
16.4.2.2 Allocation of Multiple Tasks *452*
16.4.3 Coordinating Through the Environment *453*
16.4.3.1 Construct-Ant-Solution *455*
16.4.3.2 Update-Pheromone *456*
16.4.4 Coordination Without Communication *456*
16.5 Conclusion *456*
Exercises *457*
Bibliography *459*

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17 Distributed Ledger *461*
17.1 Cryptographic Techniques *462*
17.2 Distributed Ledger Systems *464*
17.2.1 Properties of Distributed Ledger Systems *465*
17.2.2 A Framework for Distributed Ledger Systems *466*
17.3 Blockchain *467*
17.3.1 Distributed Consensus in Blockchain *468*
17.3.2 Forking *470*
17.3.3 Distributed Asset Tracking *471*
17.3.4 Byzantine Fault Tolerance and Proof of Work *472*
17.4 Other Techniques for Distributed Consensus *473*
17.4.1 Alternative Proofs *473*
17.4.2 Non-linear Data Structures *474*
17.4.2.1 Tangle *474*
17.4.2.2 Hashgraph *476*
17.5 Scripts and Smart Contracts *480*
17.6 Distributed Ledgers for Cyber-Physical Systems *483*
17.6.1 Layered Architecture *484*
17.6.2 Smart Contract in Cyber-Physical Systems *486*
17.7 Conclusion *486*
Exercises *487*
Bibliography *488*

18 Case Study *491*
18.1 Collaborative E-Learning Systems *492*
18.2 P2P E-Learning System *493*
18.2.1 Web Conferencing Versus P2P-IPS *495*
18.3 P2P Shared Whiteboard *497*
18.3.1 Repainting Shared Whiteboard *497*
18.3.2 Consistency of Board View at Peers *498*
18.4 P2P Live Streaming *500*
18.4.1 Peer Joining *500*
18.4.2 Peer Leaving *503*
18.4.3 Handling "Ask Doubt" *504*
18.5 P2P-IPS for Stored Contents *504*
18.5.1 De Bruijn Graphs for DHT Implementation *505*
18.5.2 Node Information Structure *507*
18.5.2.1 Join Example *510*
18.5.3 Leaving of Peers *510*
18.6 Searching, Sharing, and Indexing *511*
18.6.1 Pre-processing of Files *511*

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18.6.2 File Indexing *512*
18.6.3 File Lookup and Download *512*
18.7 Annotations and Discussion Forum *513*
18.7.1 Annotation Format *513*
18.7.2 Storing Annotations *514*
18.7.3 Audio and Video Annotation *514*
18.7.4 PDF Annotation *514*
18.7.5 Posts, Comments, and Announcements *514*
18.7.6 Synchronization of Posts and Comments *515*
18.7.6.1 Epidemic Dissemination *516*
18.7.6.2 Reconciliation *516*
18.8 Simulation Results *516*
18.8.1 Live Streaming and Shared Whiteboard *517*
18.8.2 De Bruijn Overlay *518*
18.9 Conclusion *520*
Bibliography *521*

Index *525*

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Preface

The concept of the book germinated about two and half years back when we were informally exchanging our experiences and thoughts on what could typically constitute a course on distributed systems for senior undergraduates and graduate students in engineering institutes. The existing textbooks on the subject, authored by eminent scientists, present an excellent discourse on the rich and interesting theoretical foundations that academicians across the board appreciate and adopt in computer science and engineering curricula. At the same time, distributed systems have recently evolved and grown beyond their conventional boundaries. There is plenty of published material on various new topics in the interfacing area of computing, communication, and Internet technologies. These developments have brought in new challenges for distributed computing technology, and we think they need to find a place in a modern curriculum on the subject. In particular, we focus on distributed applications involving large volumes of distributed data, cyber-physical systems, and distributed intelligence. Though excellent research articles and books are available on these topics, they tend to focus on aspects of the technology other than distributed computing. It is challenging to garner a holistic view of the subject by joining the knowledge dots from various books and research articles. We try to address this challenge in our work.

We debated if there is a space for a new textbook encompassing protocols, theory, and applications that also included distributed data analytics and smart environments. If so, the challenge is to organize the material and package it in a form that might have a broader academic acceptance while serving as a reference text for the developers. We drew our experiences in the roles of an instructor and a practitioner. We interacted with the students and the developers to identify the knowledge gaps that hamper their career growth in an evolving discipline. We have observed over time that the merging of communication technologies, computing, and the Internet motivated smart developers to build large applications over geographically dispersed distributed computing resources, mobile hand-held systems, and sensor-controlled smart devices. Many toolchains were developed to

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aid the building of these applications. Applications often needed to interface with human and program-controlled actors using petabytes of data stored over large data centers that communicate through the Internet. Earlier, data from different distributed sources were fed to a central computer over a telecommunication network for processing. While this approach worked satisfactorily for small and mid-sized applications, it could not scale well due to the capacity limitation of the central processing node and the excessive network traffic. Besides capacity, the reliability of the data and system availability became severe handicaps for the centralized approach. The exponential growth in data traffic due to sensory data, videos, and requirements for distributed data analytics compounded the problems for the communication networks. It was soon realized that distributed computing, where data for an application is processed on multiple independent and interconnected computers, is the key to achieving scalability and reliability in large-scale distributed application systems.

The paradigm of distributed computing has been around for several decades now with the pioneering works of Turing awardees like Leslie Lamport, Edgar W. Dijkstra, Barbara Liskov, among others. At the same time, industry requirements fueled research and development. As a result, the subject of distributed systems witnessed spectacular growth over the years. Starting with client–server applications, where some data preprocessing and rendering of the results were delegated to the client computers, distributed computing has matured to peer-to-peer systems where the participating application components could make independent decisions. We find heterogeneous devices in such peer-to-peer systems, with large servers with tremendous processing power and storage capacity at one end of the spectrum and severely constrained IoT devices at the other. We see large and "open" distributed applications, where computer systems owned by different individuals or groups, even with unknown ownership, can participate for specific periods. Addressing security concerns in such open systems created newer and non-predictable challenges to the design of distributed systems today.

In this book, we have attempted to bridge the gap between the foundational material and contemporary advances in distributed computing. To give a complete and coherent view of the subject, we start with the fundamental issues and theories of distributed computing and progressively move to advanced and contemporary topics. We present the subject in three layers of abstraction, namely,

1. *Network*, dealing with basic connectivity of the computers and processes running on them.
2. *Middleware tools*, which provide a layer of abstraction over the possibly heterogeneous network layer, and facilitates system development efforts.
3. *Application frameworks* enable the development of various distributed applications.

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In summary, we expect a reader of the book will be able to

1. Get a holistic coverage of the subject by addressing different layers of abstraction in a distributed system, namely network, middleware tools, and application framework.
2. Relate the theoretical foundations with the contemporary advances in distributed systems.
3. Familiarity with distributed computing principles deployed in the applications frameworks that are crucial for developing smart environments and distributed automation requirements for industry 4.0.

The book's content has been organized in the form of three main threads as in Figure 1. The middle thread marked "A," consisting of nine chapters, could be sufficient for the first-level course on distributed systems. An advanced level course on operating systems could consist of the first seven chapters and a few additional topics labeled by the left thread marked "B." Understanding case study requires

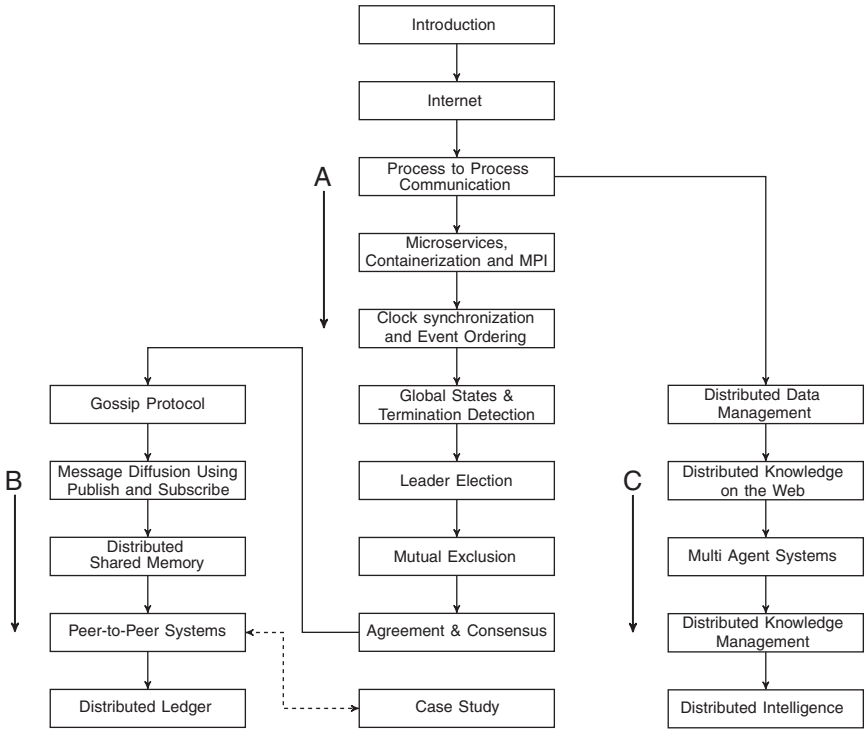


Figure 1 Topics and flow diagram of book's content.

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knowledge of peer-to-peer system apart from the basics of distributed system covered in thread "A." It is also possible to use the text for an advanced graduate-level course on distributed systems oriented toward intelligent data management and applications. It is represented by the thread marked "C" to the right in the content flow diagram.

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Acknowledgments

The subject of Distributed Computing has seen fascinating growth over the last couple of decades. It has resulted in many practical and useful systems that pervade our daily lives. At the outset, we acknowledge the efforts of numerous researchers and practitioners, without which the content of this book would not have materialized.

This book grew from several courses on Distributed Systems, and related topics offered to senior undergraduate and graduate students at many institutes, namely IIT-Kanpur, IIT-Bhilai, IIT-Delhi, and IIT-Jodhpur. The interaction with the students and the experiments conducted with their help were an incredible learning process. We acknowledge the contributions of these students to shaping the book's contents. We also acknowledge the encouragement of our colleagues at these institutes, who contributed valuable inputs to defining the curricula for the subjects.

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Acronyms

2PC	two phase commit
3PC	three phase commit
6LowPAN	IPv6 over low-power wireless personal area networks
ACL	agent communication language
ACO	ant colony optimization
AMQP	advanced message queuing protocol
AMS	agent management services
API	application programming interface
AS	autonomous system
ASIC	application specific integrated circuit
BBB	BigBlueButton
BFT	Byzantine fault-tolerance
BGP	Byzantine general problem
BGP	basic graph pattern
BIRCH	balanced iterative reducing and clustering using hierarchies
BM	block manager
BSS	Birman Stephenson and Schiper
BTC	bitcoin
C	consensus
CAN	content addressable P2P network
CDN	content distribution network
CDPS	cooperative distributed problem solving
CFP	call for proposal
CGI	common gateway interface
CNAME	canonical name
CNP	contract net protocol
CoAP	constrained application protocol
CoTS	components of the shelf
CPU	central processing unit

CRUD	create, read, update(/write), delete
CS	critical section
CSMA/CA	carrier sensing multiple access with collision avoidance
CSMA/CD	carrier sensing multiple access with collision detection
CSV	comma-separated values
CTP	collection tree protocol
DAG	directed acyclic graph
DARPA	defense advanced research projects agency
DCMI	Dublin core metadata initiative
DDBMS	distributed data base management
DFS	distributed file systems
DHCP	dynamic host control protocol
DHT	distributed hash table
DLT	distributed ledger technology
DNS	domain name service
DnS	descriptions and situations
DOI	digital object identifier
DOLCE	descriptive ontology for linguistic and cognitive engineering
DSN	distributed sensor network
DUL	DOLCE+DnS ultralite
ECDSA	elliptic curve digital signature algorithm
ETX	expected transmission count
FCFS	fist come first serve
FELGossip	fair efficient location-based gossip
FIFO	first in first out
FiGo	firefly gossip
FIPA	Foundation of Physical Agents
FoaF	friend of a friend
FPGA	field programmable gate array
FQDN	fully qualified domain name
FT	finger table
FTP	file transfer protocol
GALS	globally asynchronous and locally synchronous
GB	giga bytes (10^9 bytes)
Gbps	giga-bits per second (10^9 bits per second)
GHS	Gallagher Humblet and Spira
GID	group ID
GMT	Greenwich mean time
GPS	global positioning system
HDFS	hadoop distributed file system
HMR	home mediation router

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HPC	high performance computing
HTML	hypertext markup language
HTTP	hypertext transfer protocol
IC	interactive consistent
ICANN	international committee for assigned names and numbers
IDL	interface definition language
IEC	International Electrotechnical Commission
IEEE	Institution of Electrical and Electronics Engineers
IoT	internet of things
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
IRI	international resource identifier
ISIS	intermediate system to intermediate system
ISO	International Standard Organization
ISP	internet service provider
IST	Indian Standard Time
JADE	Java agent development environment
JPEG	joint photographic experts group
JSON	JavaScript Object Notation
JVM	Java virtual machine
KQML	Knowledge Query and Manipulation Language
LAN	local area network
LCA	lowest common ancestor
LCP	lowest common prefix
LGossip	location-based gossip
LLN	low power lossy network
LM	local manager
M2M	machine to machine
MAC	media access control
MAS	multi-agent system
MFENCE	memory fence
MMU	memory management unit
MOC	message oriented communication
MPI	message passing interface
MQTT	message queue telemetry transport
MR	mediation router
MRMW	multiple readers, multiple writers
MRSW	multiple reader, single writer
MSB	most significant bit
MTU	maximum transmission unit

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NAT	network address translation
NIST	National Institution of Standard and Technology
NoC	network on chip
NoSQL	not (only) SQL
NS	name server
NTP	network time protocol
NUMA	non uniform memory access
OASIS	Organization for the Advancement of Structured Information Standards
OM	oral message
OS	operating systems
OWL	web ontology language
P2P	peer to peer
P2P-IPS	peer to peer interactive presentation system
PDF	portable document format
PGM	property graph model
PHP	hypertext preprocessor
PID	process ID
PoET	proof of elapsed time
PoS	proof of stake
POSIX	portable operating system interface
PoW	proof of work
PSO	partial store order
PTR	pointer record
QoS	quality of service
RAID	redundant array of inexpensive disks
RDF	resource description framework
RDFS	RDF schema
REST	REpresentational State Transfer
RFC	request for comments
RMI	remote method invocation
RNS	root name server
RPC	remote procedure call
RPS	random peer sampling
RTT	round trip time
S-DSM	software distributed shared memory
SAN	storage area network
SASL	simple authentication and security layer
SC	sequential consistency
SDSS	Sloan digital sky survey
SES	Schiper Eggli and Sandoz

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SHA	secure hash algorithm
SI	susceptible and infected
SIP	session initiation protocol
SIR	susceptible, infected and removed
SKOS	simple knowledge organization system
SMP	symmetric multi-processor
SMR	shingled magnetic recording
SNS	social network systems
SOA	service oriented architecture
SOAP	simple object access protocol
SPARQL	SPARQL protocol and RDF query language
SQL	structured query language
SRSW	single reader single writer
SSH	secured shell
SSN	semantic sensor network
TB	terra bytes (10^{12} bytes)
TCP	transport control protocol
TDB	triplet data base
TF-IDF	term frequency-inverse document frequency
TSO	total store order
TTL	time to live
UDP	user datagram protocol
UMA	uniform memory access
URI	universal resource identifier
URL	uniform resource locator
UT	universal time
UTC	coordinated universal time
VM	virtual machine
W3C	world wide web consortium
WAN	wide area network
WSDL	web services description language
WSGI	web server gateway interface
WSN	wireless sensor network
XML	extensible markup language
XQuery	XML query language

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